Worksheet 6-8 & 6-7 C Basic Integrals and the FTC #2
AP Calculus Name_____

Directions: Evaluate the following integrals

1. $\int (3x+4)^2 \, dx$

2. $\int \left(3x^2 + 3\sin x + \frac{1}{x}\right) \, dx$

3. $\int \left(4x^3 + \csc^2 x + \frac{2}{x^2}\right) \, dx$

4. $\int (\cos x + \sec^2 x - 8x^5) \, dx$

5. $\int \left(7x^3 - \frac{4}{5}x^{-9/2} + x^{4/3}\right) \, dx$

6. $\int (x^{1/2} + 3x^2) \, dx$

7. $\int \left(\frac{2x^2}{\sqrt{x}} + \frac{3\sqrt{x}}{7x^2}\right) dx$

8. $\int \left(\frac{8}{\sqrt{x}} + 7\sqrt{x}\right) dx$

9. $\int \left(\frac{x}{\sqrt{x^3}} + \frac{4x^3}{3} + \frac{\sqrt[3]{x}}{2}\right) dx$

Directions: Evaluate the following integrals. Give the exact answer.

10. $\int_0^2 x^3 \, dx$

11. $\int_{\pi/6}^{\pi/2} \sqrt{3} \sin x \, dx$

12. $\int_{-1}^{1} 6x^2 \, dx$

13. $\int_0^{\pi/3} \sec x \tan x \, dx$

14. $\int_1^2 \left(2x^3 + 2x + \sqrt{x}\right) \, dx$

15. $\int_1^2 \left(x^4 - \frac{7}{x^3} + \frac{5}{\sqrt{x}}\right) \, dx$